

[121]

SEAT No. _____

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SARDAR PATEL UNIVERSITY

M. Sc. (IT) INTEGRATED EXAMINATION, VIII SEM

Date : 23rd Mar 2019, Saturday

Session : Evening

Course No : PS08CIIT03

Time : 02:00pm To 05:00pm

Sub: Artificial Intelligence

Total marks : 70

Q - 1 Select an appropriate option

[08]

- i) From the following which is not activity of KBS Development model .
 - a) Knowledge acquisition
 - b) Feasible requirement
 - a) c) System Development
 - d) None of above
- ii) What is Artificial intelligence?
 - a) Making a machine intelligent
 - b) Programming with your own intelligence
 - c) Putting your intelligence into computer
 - d) Putting your memory into computer
- iii) Factual knowledge uses _____ symbol for 'implies'.
 - a) ^
 - b) v
 - c) ~
 - d) ==>
- iv) CLIPS stands for.
 - a) C Language Integrated Production System
 - b) C Language Interactive Purchasing System
 - c) C Language Inference Production System
 - a) d) None of Above
- v) The power and strength of fuzzy logic comes from its ability to deal with
 - a) a) Membership function
 - b) Fuzzy set
 - b) c) Membership function
 - d) Linguistic variables
- vi) In fuzzy connectives implications are obtained by _____.
 - a) Negation & Intersection
 - b) Intersection
 - c) Union
 - d) Negation
- vii) _____ is a collection of such genes representing possible solutions in domain space.
 - a) Genetic Cycle
 - b) Genetic algorithm
 - c) Mutation
 - d) Genotype
- viii) Genes that are not strong are combined with their mates and used to generate new individuals in a process known as _____.
 - a) Recombination
 - b) crossover
 - c) Mutation
 - d) genotype

Q - 2 Answer the following Questions (Any SEVEN)

[14]

- i) Write any two definition of AI.
- ii) What is the roll of knowledge in decision making?.
- iii) List four reasons for solving complex problem using AI technique.
- iv) Draw a diagram of development methodology for a Knowledge Based System.
- v) List steps for knowledge acquisition.
- vi) Define Fuzzification and Defuzzification.
- vii) Write probabilistics Neural Networks.
- viii) Draw a figure of Genetic Cycle.
- ix) Write any six application and research areas where Genetic algorithms can apply

Q - 3 a) Write note on inference engine.

[06]

b) Explain Applications of Knowledge Based System.

[06]

OR

Q - 3 b) Write difference between KBS and CBIS

[06]

- Q - 4 (a) Explain existing techniques for Knowledge aquisition. [06]
(b) Explain Knowledge based system tools. [06]
OR
- Q - 4 (b) Explain factual knowledge. [06]
- Q - 5 a) Explain Fuzzy control system in detail. [06]
b) Explain Fuzzy propositions and inference. [06]
OR
- Q - 5 b) Explain the neural network architecture. [06]
- Q - 6 a) Write introduction and basic operators of a Genetic Algorithm [06]
b) Write about application and research trends in Genetic Algorithm. [06]
OR
- Q - 6 b) Minimize a simple function $f(x)=x*x$ on the integer interval $[0,30]$ using genetic algorithm. Take the 01101, 11000, 01000, 10011 strings as individuals in the first population [06]

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(2)